DEEPMATH 2026 / ACCEPTED CONFERENCE POSTER

Weight-Sharing Order Changes Linear Expressivity

Separation and Commutator Bounds



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01 Same weights. Different order.

Can reuse order change what a linear network can represent?

ALTERNATING
Execution order



$$T_{ABAB} = BABA = (BA)^2$$

BLOCK
Execution order



$$T_{AABB} = B^2A^2$$

2 real matrices · **4** layers · **(2, 2)** reuse frequency · **2n²** parameters

02 A strict expressivity separation

For real n × n matrices, n ≥ 2, the block family is strictly larger.

$$\{(BA)^2\} \subsetneq \{B^2A^2\}$$

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},\ B = \mathrm{diag}(1, 2),\ B^2A^2 = \mathrm{diag}(-1, -4)$$

1 Inclusion

Every alternating map is a real matrix square. Every real matrix square is also a block map: take A = I.

2 A witness

If $C^2 = D = \mathrm{diag}(-1, -4)$, then C commutes with D and preserves its one-dimensional −1 eigenspace.

3 An obstruction

On that eigenspace, a real scalar would have to square to −1. Hence D has no real square root.

Extend A and B with identity blocks for n > 2. **The separation holds even with invertible weights.**

03 Evidence, with the limits in view

MNIST + Fashion-MNIST: 60,000 train / 10,000 test each. Training-only PCA: n = 16, 32.
Four schedules × ten seeds = 160 fits. AABB / ABAB use matched initial weights.

Test accuracy (%) · mean ± sample SD, ten seeds

Dataset	n	AABB	ABAB
MNIST	16	77.550 ± 0.000	77.556 ± 0.008
MNIST	32	83.600 ± 0.000	83.560 ± 0.000
Fashion-MNIST	16	71.520 ± 0.000	72.008 ± 0.033
Fashion-MNIST	32	75.960 ± 0.000	76.034 ± 0.011

Bias-free linear maps · L-BFGS-B · MSE + $10^{-4}\|T\|^2F$ penalty

1,200 dependent comparisons

80 learned models × 15 pairs of reorderings

77.5% (930 / 1,200) — **Uniform bound**
Improves norm-only baseline $2M^4$

7.7% (92 / 1,200) — **Transported bound**
Improves tighter baseline $\|T\sigma\| + \|T\tau\|$

Different baselines; descriptive, dependent comparisons.

80 / 160 fits hit the 10,000-iteration cap. All AAAA and ABAB fits hit the cap; all AABB and ABCD fits met stopping criteria. These are finite-budget outcomes, not a demonstrated classification advantage.

04 Sensitivity to order

For fixed weights, an adjacent swap contributes a commutator scaled by the surrounding factors.

$$\Delta T = P[W_j, W_i]Q$$
$$[A, B] = AB - BA$$

TRANSPORTED BOUND

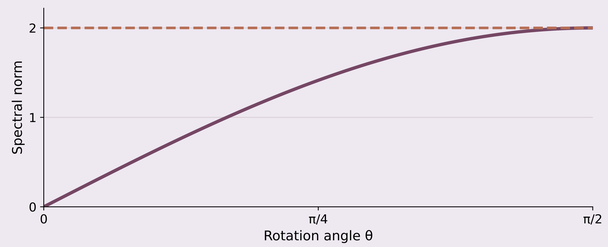
$$\|T_\sigma - T_\tau\| \le \min_\gamma \sum_{r\in\gamma} \|P_r\|\,\|[W_{j_r}, W_{i_r}]\|\,\|Q_r\|$$

UNIFORM BOUND

$$\|T_\sigma - T_\tau\| \le M^{L-2} C_W(\sigma, \tau)$$

M is the maximum weight norm; C_W is the minimum sum of commutator norms along adjacent-swap paths. All norms are spectral.

An exactly attainable regime



Analytic example: A = diag(1, −1), B = R(θ), identity exterior factors. Both commutator bounds equal 2|sin θ|; the norm-only bound is 2.

Different maps, same Gram geometry

With orthogonal factors, every ordering gives:

$$X^\top T_\sigma^\top T_\sigma X = X^\top X$$

Sharing order is an architectural choice.

Representability, sensitivity, and predictive performance are separate questions. Optimization failure is not proof of nonrepresentability; claims about nonlinear LLMs require separate evidence.

SELECTED REFERENCES

Takase & Kiyono (2023), SustaiNLP, pp. 78-90.
Gallier, Logarithms and square roots of real matrices, arXiv:0805.0245.
Childs et al. (2021), Physical Review X 11:011020.